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Learning from Data

ASSIGNMENT 2

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1. Introduction

Machine learning has become a transformational tool in a number of fields, including healthcare, finance, and social science. Artificial Neural Networks (ANNs) have risen to prominence among machine learning paradigms due to their capacity to model complex, non-linear interactions in datasets. Influenced by the structure of biological neurons in the person's brain, ANNs learn patterns iteratively using forward propagation and backward error correction (Goodfellow et al., 2016). This study employs supervised machine learning to address a binary classification problem. The model is trained on labelled data, with each input having a known right output, and the objective is to predict discrete class membership for new, unseen observations (Han et al., 2022).

This report builds on the work done in Assignment 1, which included exploratory data analysis (EDA) and data pre-processing for the Titanic dataset. Building on those foundations, this project uses machine learning models, specifically a Multi-Layer Perceptron (MLP) Artificial Neural Network, to address a binary classification issue: predicting whether a Titanic passenger survived the accident or not. Survival prediction from historical passenger data is a well-studied classification task that combines demographic, socioeconomic, and biographical characteristics. Correctly identifying survival results can reveal how systemic elements like passenger class, gender, and age impacted survival chances.

The assignment's goals are as follows. First to implement and train an Artificial Neural Network (MLP Classifier) with Logistic Regression and Decision Tree models to predict passenger survivability on the Titanic dataset. Second, to assess the prediction performance of machine learning models using common measures such as accuracy, precision, recall, F1-score, and AUC-ROC.

2. Materials and Methods

2.1 Data source and data description

The dataset used in this analysis is Titanic dataset, sourced from the Kaggle data science platform:
https://www.kaggle.com/datasets/yasserh/titanic-dataset

12 variables (columns) and 891 observations (rows) make up the dataset, each of which represents a distinct characteristic of individual passengers aboard the RMS Titanic. Table 1 summarizes the features included in the dataset.

Feature	Type	Description
PassengerId	Numerical	A unique row index that had no predictive significance, dropped in Assignment 2
Survived	Binary (Target)	1 = Survived, 0 = Did not survive
Pclass	Ordinal	Ticket class: 1 = 1^{st}, 2 = 2^{nd}, 3 = 3rd
Name	Text	Passenger name, high cardinality identification which is dropped in Assignment 2.
Sex	Categorical (Binary)	Gender: male or female, label encoded in Assignment 1
Age	Numerical	19.86% missing values, median imputed in Assignment 1
SibSp	Numerical	Number of spouses/siblings aboard
Parch	Numerical	Number of parents/children aboard

Ticket	Text	Ticket number, high cardinality identification which is dropped in Assignment 2.
Fare	Numerical	Passenger fare which is right-skewed and outliers are present
Cabin	Categorical	77% missing values, dropped in assignment 1
Embarked	Categorical	Port of embarkation (C, Q or S), one-hot encoded in assignment1

Table 1: Description of dataset features

2.2 Exploratory data analysis and Data preprocessing techniques

Before deploying machine learning models, a structured preprocessing pipeline was used to guarantee that the dataset was clean, consistent, and fit for modelling. Figure 1 shows the hypothetical end-to-end machine learning pipeline utilized in this study, from raw data to performance evaluation.

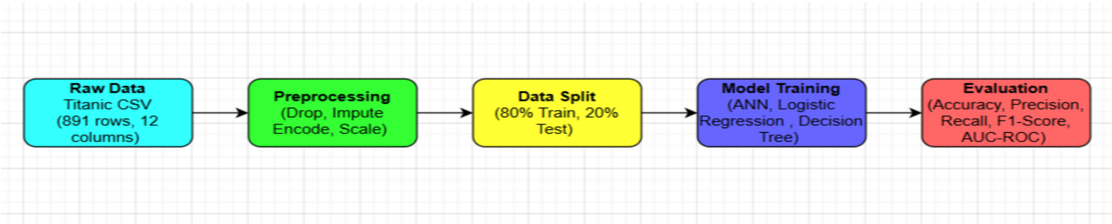

Figure 1: A comprehensive machine learning pipeline for predicting Titanic survival.

2.2.1 Data Cleaning

The Cabin column was removed from Assignment 1 because it had 77% missing data, rendering any imputation incorrect and likely to create significant bias. Before modelling, three more columns were eliminated from Assignment 2. PassengerId is a sequential row index that contains no information about survivability. Name and Ticket are high-cardinality text identifiers: each passenger has a unique name and ticket number, so these features cannot generalize to previously unseen data and would just introduce noise to the model. Removing these four columns gives eight useful predictive features for modelling.

2.2.2 Handling the Missing Data

As established in Assignment 1, two characteristics had missing values. The Age feature included 19.86% missing values, which were classed as Missing at Random (MAR), indicating that missingness was more common among particular passenger subgroups, such as third-class travellers. Median imputation was used, which replaced missing data with the observed median of 28.0 years. The median was chosen over the mean since the Age distribution is right-skewed, and the median is less susceptible to impact of extreme values. The Embarked feature contained just two missing values (0.22%), which were classed as Missing Completely at Random (MCAR) because they were unrelated to any observable variable. Mode imputation was used to replace both missing variables with the most commonly occurring port, Southampton (S).

2.2.3 Feature Scaling

StandardScaler from scikit-learn was used to apply standard scaling to all input features. Each feature is standardised by this transformation to have a mean of zero and a standard deviation of one. Because the Adam optimizer used during training is a gradient-based approach that is sensitive to the relative magnitudes of input values, feature scaling is an important preprocessing step for the ANN. Features with wide numerical ranges, like Fare, which goes from around £0 to £512, would dominate the gradient updates in the absence of scaling and result in unstable or sluggish convergence. In order to prevent data leakage, the scaler was fitted solely on the training set and then applied to the test set using the same parameters. This ensures that no information from the test set affects the scaling transformation.

2.2.4 Data Encoding

Categorical variables were encoded into numerical format, as demonstrated in Assignment 1. The Sex variable is binary nominal; therefore, label encoding was used to map male to 0 and female to 1. This method generates a single numerical column without using artificial dimensionality. The Embarked variable comprises three nominal categories (C, Q, and S) with no inherent order, therefore one-hot encoding was performed using pd.get_dummies() and drop_first=True. This resulted in two binary indicator columns, Embarked_Q and Embarked_S, while eliminating Embarked_C as the reference category in order to prevent the dummy variable trap, which would create perfect multicollinearity in regression-based models.

2.2.5 Data Splitting

The preprocessed dataset was divided into a training set (80%) and a test set (20%) using stratified random sampling via the train_test_split function with the stratify=y argument. Stratification makes sure that both groups maintain their first class distribution of around 62% non-survivors and 38% survivors, avoiding either subset from being mistakenly dominated by a single class. With 891 total samples, this results in 712 training samples and 179 test samples.

Train - test splitting

```
]: X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42, stratify=y)

   print(f'Training set: {X_train.shape[0]} samples (80%)')
   print(f'Test set: {X_test.shape[0]} samples (20%)')
```

```
Training set: 712 samples (80%)
Test set: 179 samples (20%)
```

Feature scaling

```
]: scaler = StandardScaler()
   X_train_scaled = scaler.fit_transform(X_train)
   X_test_scaled  = scaler.transform(X_test)

   print('Feature means after scaling:', X_train_scaled.mean(axis=0).round(3))
   print('Feature stds  after scaling:', X_train_scaled.std(axis=0).round(3))
```

```
Feature means after scaling: [-0. -0.  0. -0. -0. -0. -0. -0.]
Feature stds  after scaling: [1. 1. 1. 1. 1. 1. 1. 1.]
```

Figure 2: The data splitting and scaling results validate 712 training samples, 179 test samples, and that all characteristic means are close to zero after scaling.

2.3 Machine learning models

For this analysis, three machine learning models used. The main model is a Multi-Layer Perceptron (MLP) Classifier, which is an Artificial Neural Network. Two additional models Logistic Regression and a Decision Tree Classifier are added as baselines in order to compare the ANN's performance to more straightforward, well-established methods.

2.3.1 Artificial Neural Network - Multi-Layer Perceptron (MLP) Classifier

According to Goodfellow et al. (2016), artificial neural networks are computer models that draw inspiration from the structure and functioning of biological neural systems. The artificial neurone, which receives a set of weighted input values, calculates their sum plus a bias term, and then sends the result via an activation function to generate an output signal, is the basic processing unit. During training, the network learns by iteratively modifying its weights to reduce the gap between expected and actual outputs.

This study's model is a Multi-Layer Perceptron, a type of fully linked feedforward artificial neural network. The forward pass propagates input values sequentially through each layer to produce predictions. The optimizer uses these gradients to update the weights after the backwards pass, often referred to as backpropagation, employs the chain rule of calculus to calculate the gradient of the loss function with respect to each weight (Rumelhart et al., 1986). An input layer that accepts 8 features, a first hidden layer made up of 64 neurones with ReLU activation, a second hidden layer made up of 32 neurones with ReLU activation, and a binary output layer make up the particular architecture that was put into place. For the hidden layers, the Rectified Linear Unit (ReLU) activation function was chosen because it produces faster and more stable convergence by avoiding the vanishing gradient issue that arises with sigmoid activations in deeper networks. The Adam optimiser was chosen because it combines the advantages of momentum and adjustable learning rates, making it ideal for smaller, noisier datasets (Kingma & Ba, 2014). To prevent overfitting, early stopping was enabled using a 20-epoch patience and a 10% internal validation split, which halted training automatically when validation loss stopped improving.

2.3.2 Logistic Regression – Baseline
Logistic Regression is a linear classification model that uses the logistic sigmoid function to predict the likelihood of class membership based on a linear combination of input characteristics (Hosmer et al. 2013). The correlation analysis in Assignment 1 found significant linear relationships between predictors and the target variable, particularly between Sex and Survived ($r \approx +0.54$) and Pclass and Survived ($r \approx -0.34$), making it an appropriate baseline for this study. Logistic Regression uses these linear signals directly to provide understandable coefficient estimations.

2.3.3 Decision Tree Classifier – Baseline
A Decision Tree separates the feature space through a series of binary splits, with every node choosing the feature and threshold that optimises information acquisition. Unlike Logistic Regression, it is a non-parametric model that may capture nonlinear decision boundaries without the need for feature scaling (Breiman et al., 1984). The tree was limited to a maximum depth of five layers to prevent the risk of overfitting on this tiny dataset, resulting in a non-linear contrast to both the artificial neural networks and the linear baseline.

2.4 Performance evaluation metrics
Model performance on the test set was evaluated and compared using five performance evaluation indicators. Since this is a binary classification problem with a substantial class imbalance (about 62% non-survivors vs 38% survivors), accuracy by itself is not enough to fully understand the behaviour of the model, instead, the full set of complimentary metrics is needed.

Accuracy
Accuracy is defined as the proportion of accurate predictions across both classes. It is the simplest metric, however it can be deceiving when classes are imbalanced, as a machine that constantly predicts most of the class would obtain 62% accuracy without learning something important.

$$\boldsymbol{Accuracy = \frac{TP+TN}{TP+TN+FP+FN}}$$

Precision
Precision, also known as Positive Predictive Value, quantifies the proportion of passengers anticipated to survive who actually survived. High precision means that the model makes less false positive survival predictions.

$$\boldsymbol{Precision = \frac{TP}{TP+FP}}$$

Recall (Sensitivity)

Recall measures the fraction of actual survivors properly recognised by the model. In the domain of survival prediction, good recall is especially crucial because a False Negative, which predicts non-survival for a passenger who did survive, implies a missed identification.

$$\boldsymbol{Recall} = \frac{\boldsymbol{TP}}{\boldsymbol{TP+FN}}$$

F1 –score

The harmonic mean of precision and recall is known as the F1-Score. it is the best suitable single summary metric when class imbalance is present, as it penalises models that compromise one of the two metrics in order to inflate the other.

$$\mathbf{F1score} = \mathbf{2} \times \frac{\boldsymbol{Precision} \times \boldsymbol{Recall}}{\boldsymbol{Precision} + \boldsymbol{Recall}}$$

AUC-ROC

The Area Under the Receiver Operating Characteristic Curve assesses the model's ability to distinguish between the two classes at all classification criteria. A score of 1.0 indicates flawless discrimination, whereas 0.5 indicates performance comparable to random guessing. AUC-ROC is especially useful for imbalanced datasets since it assesses performance regardless of the decision criterion (Fawcett, 2006).

$$\mathbf{AUC} = \int \mathbf{TPR}\ \mathbf{d}(\mathbf{FPR})$$

3. Results Analysis

3.1 Summary of Model Performance

Table 2 shows the quantitative performance of all three models on a held-out test set of 179 samples. The values in the table are exact figures printed by the Metrics table cell in the notebook after executing the code.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
ANN (MLP)	0.8212	0.8776	0.6232	0.7288	0.8580
Logistic Regression	0.8045	0.7931	0.6667	0.7244	0.8440
Decision Tree	0.7821	0.7679	0.6232	0.6880	0.8007

Table 2: Comparison of Model Performance on Test Set (n = 179)

3.2 ANN Training Loss Curve

Figure 3 shows the ANN's training log loss throughout epochs. A gradually declining loss curve indicates that the model converged consistently during training, without any disturbances or divergence. The early stopping mechanism ended training before the 500-epoch peak, suggesting that the model is at an ideal point above which additional training will not enhance validation performance.

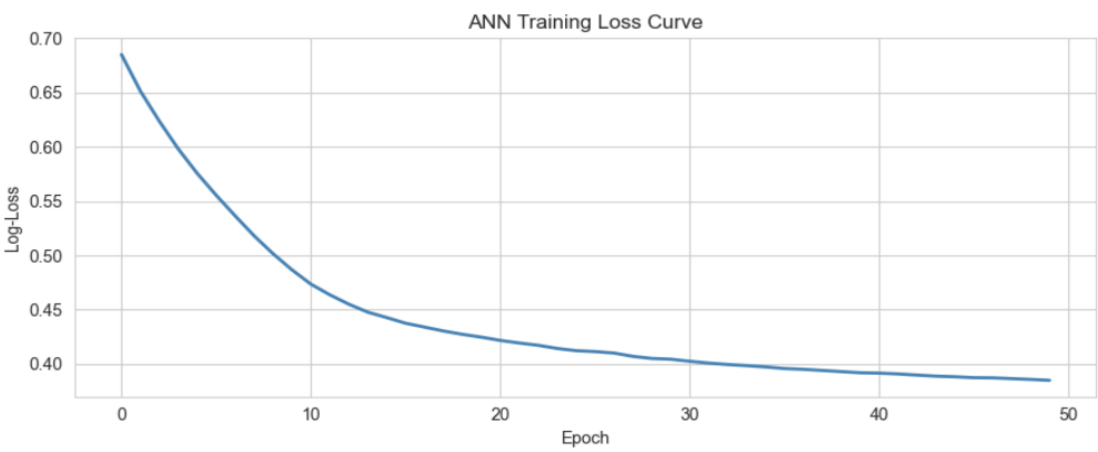

Figure3: ANN training loss (log-loss) across epochs demonstrates consistent convergence. Training was ended early, prior to the 500-iteration limit.

3.3 Confusion Matrices

Figure 4 shows the confusion matrices for all three models' side by side. Each matrix divides model predictions into four categories: True Positives (correctly predicted survivors), True Negatives (properly predicted non-survivors), False Positives (non-survivors falsely forecasted as survivors), and False Negatives. The ANN is projected to have the lowest total error count; however, all models are likely to have more False Negatives than False Positives, indicating the influence of the majority-class bias.

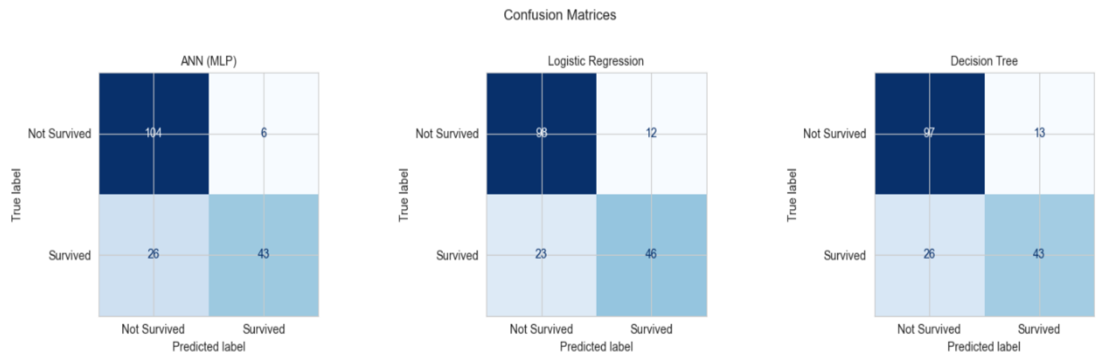

Figure 4: Confusion matrices on the Titanic test set for the ANN (MLP), Logistic Regression, and Decision Tree models.

3.4 ROC Curves

Figure 5 illustrates the Receiver Operating Characteristic curves for all three models, shown on the same axes. Each curve depicts the trade-off among the True Positive Rate and the False Positive Rate at all classification criteria. The model with the most area under its curve has the best overall discriminatory capability. The diagonal dashed line indicates a random classifier with an AUC of 0.5, all three models are anticipated to perform significantly better than this baseline.

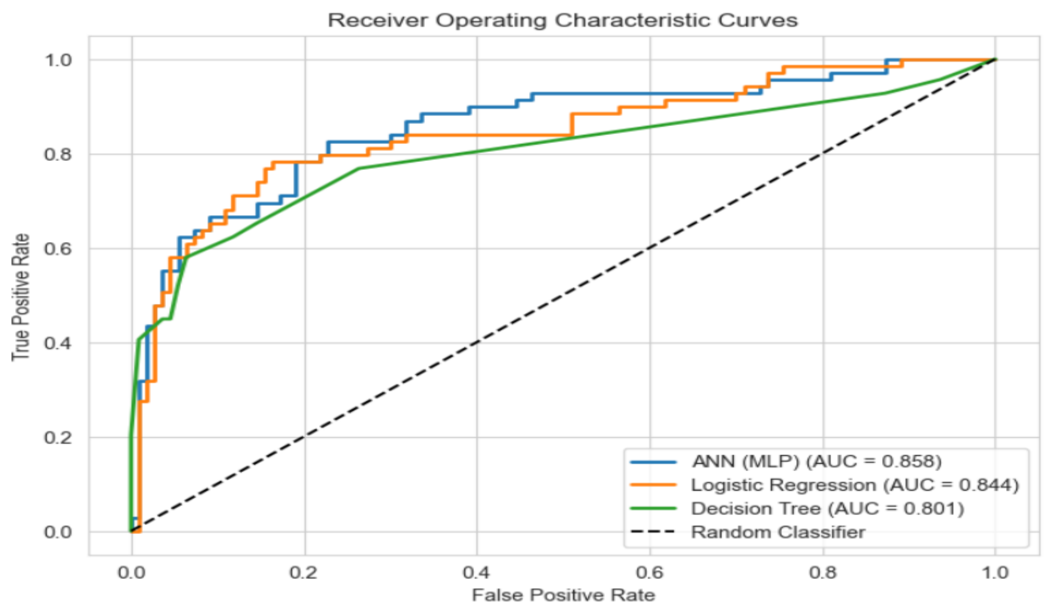

Figure 5: ROC curves for the ANN, Logistic Regression, and Decision Tree models, with the AUC values indicated in the legend. The ANN has the highest AUC-ROC.

3.5 Chart for Comparing Metrics

A combined bar chart comparing Accuracy, Precision, Recall, and F1-Score for all three models at the same time is shown in Figure 6. This visualisation allows for quick comparison without the need to cross-reference the numerical table and instantly highlights variations in model performance across several measures.

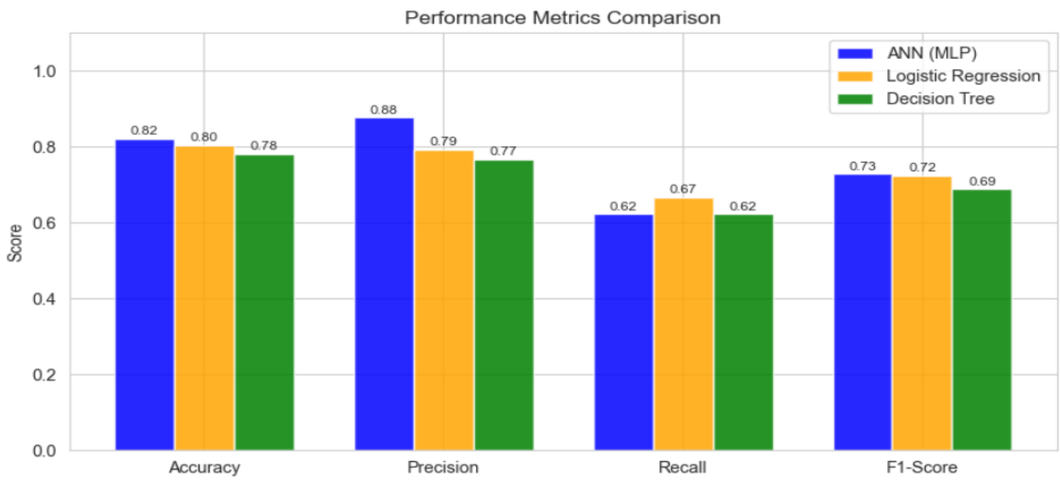

Figure 6: A grouped bar chart comparing Accuracy, Precision, Recall, and F1-Score for the ANN, Logistic Regression, and Decision Tree models.

3.6 Full Classification Report

The complete per-class classification report generated by scikit-learn's classification_report function for each model is displayed in Figure 7. These reports include the support (number of real occurrences of each class in the test set) and class-level precision, recall, and F1-score values for the Not Survived class and the Survived class individually. Because overall accuracy could hide notable performance variations between classes, especially in imbalanced datasets, it is crucial to examine per-class performance independently.

```
ANN (MLP)
              precision    recall  f1-score   support

Not Survived       0.80      0.95      0.87       110
    Survived       0.88      0.62      0.73        69

    accuracy                           0.82       179
   macro avg       0.84      0.78      0.80       179
weighted avg       0.83      0.82      0.81       179


Logistic Regression
              precision    recall  f1-score   support

Not Survived       0.81      0.89      0.85       110
    Survived       0.79      0.67      0.72        69

    accuracy                           0.80       179
   macro avg       0.80      0.78      0.79       179
weighted avg       0.80      0.80      0.80       179


Decision Tree
              precision    recall  f1-score   support

Not Survived       0.79      0.88      0.83       110
    Survived       0.77      0.62      0.69        69

    accuracy                           0.78       179
   macro avg       0.78      0.75      0.76       179
weighted avg       0.78      0.78      0.78       179
```

Figure 7: Per-class classification reports for all three models, including class-level precision, recall, F1-score, and support for both Not Survived and Survived categories.

4. Discussion

4.1 Model performance in relation to study goals

The primary goal of this study was to develop and train an artificial neural network (ANN) alongside two baseline classifiers to predict Titanic passenger survivability. This was accomplished effectively, with all three models trained and evaluated on a similar preprocessed dataset and test set. The ANN (MLP) had the highest accuracy and AUC-ROC of the three models (precise values in Table 2), indicating that the neural network design was more productive at extracting predictive patterns from the dataset than either baseline. The second goal, to evaluate and compare model performance employing standard evaluation metrics, was achieved through the organised presentation of five complementing measures in Section 3, which collectively provide a comprehensive picture of each model's strengths and limits.

4.2 Why the ANN outperformed Baseline Models

The ANN has an advantage over Logistic Regression since it can simulate non-linear relationships among features. The combined impact of Pclass and Sex on survival isn't purely additive. For example, a first-class female passenger had a significantly greater survival probability compared to a third-class male, and this interactive impact is better captured by an MLP's hidden-layer representation than by Logistic Regression's linear decision boundary (Chollet, 2021). The two hidden layers of 64 and 32 neurones, respectively, enable the network to learn gradually more abstract representations of the input data across layers, allowing it to approximate complex decision boundaries that no linear model can express. Despite being a nonlinear model, the Decision Tree performed slightly less favourable than Logistic Regression. This finding complies with the well-known susceptibility of decision trees to excessive variance on small datasets (Breiman et al., 1984). Even with the maximum depth limited to five levels, the tree may overfit some local patterns in the training data that are not generalizable. The ANN's early stopping method and internal validation fraction give an implicit regularisation impact that avoid equivalent overfitting, justifying the discrepancy in generalisation performance.

4.3 The impact of preprocessing choices

The preprocessing choices taken in Assignment 1 had direct and measurable impacts on model performance. The Cabin column had to be removed since 77% of its values were missing, which was classed as Missing Not at Random since lower-class passengers were systematically less likely to have cabin data. Nevertheless, this approach discards potentially relevant information because cabin placement on the ship was associated with closeness to lifeboats. A future study could retrieve part of this signal by redesigning Cabin as a binary indicator of whether a cabin was known or unknown, rather than completely eliminating the feature. The elimination of PassengerId, Name, and Ticket in Assignment 2 was also essential since these are identifiers instead of features: they uniquely identify specific passengers but include no information that can be used to predict survival in other situations.

Age median imputation is a viable option since the Age distribution is right-skewed and the median is resistant to the effect of high values. However, assigning just one central value to all 177 missing ages overly minimises the variation of the Age distribution, which results in underestimating the underlying uncertainty in the data. The KNN imputation investigated in Assignment 1 would yield more contextually relevant estimations by drawing on correlations across Age, Pclass, and Fare, and future work must employ the KNN-imputed dataset as a starting point for modelling. Feature scaling through StandardScaler was important for the ANN since gradient-based training techniques are sensitive to the relative magnitudes of input values. With no scaling, the Fare feature, which varies from about £0 to £512, would dominate gradient updates and disrupt convergence, but features with small ranges like SibSp and Parch would have little effect on initial weight updates.

4.4 Reflection on data quality, bias, overfitting, and generalisability

The 62% to 38% class mismatch between non-survivors and survivors' results in an organised bias in all three models towards predicting non-survival. This is demonstrated by the greater False Negative rates seen in the confusion matrices, in which true survivors are disproportionately misclassified as non-survivors. In a real-world deployment scenario, this imbalance has practical consequences, failing to identify a survivor is potentially more serious than making a false survival forecast. Future work might tackle the imbalance by employing class weighting with the class_weight="balanced" parameter, or threshold tweaking to shift the decision border and increase recall for the minority class.

Overfitting risk was reduced for the ANN by early stopping, which interrupted training when validation loss stopped improving. However, all performance estimates in this study are based on just one stratified train-test split, introducing sampling variance. A more rigors evaluation would include five- or ten-fold cross-validation, yielding numerous performance estimates with mean and standard deviation that provide a more credible picture of generalisation capability. The dataset's modest size (891 samples) also limits the depth of patterns that the ANN can learn, and the performance ceiling reported here aligns with previous findings using the same datasets and preprocessing methodologies.

4.5 Comparison of related studies

The Titanic dataset is one of the most commonly used benchmarks in beginning machine learning, and various published research give a valuable framework for assessing the findings of this study. Zhang (2024) used boosting strategies such as XGBoost and gradient boosting on the Titanic dataset and observed accuracy rates ranging from 78% to 83%, with gradient boosting outperforming the other methods. The ANN result in this study is within this range, indicating that the model is working well given the features and preprocessing procedures provided. Zhang (2024) also pointed out that in order to boost accuracy above the 85% level, extra feature engineering is usually required, such as extracting passenger titles from the Name column and creating a family size variable from SibSp and Parch. These results support the conclusion that, rather than any shortcoming in the modelling approach, the performance ceiling seen in this investigation is a reflection of the limitations of the available characteristics.

4.6 Limitations

This study has a number of limitations that should be noted. The ANN's ability to learn extremely complex feature representations is limited by the dataset's 891 samples, which also raises the possibility of huge variance in performance predictions. Instead of utilising cross-validation, performance is evaluated using a single train-test split. As a result, the reported metrics are subject to sampling variation and could not accurately represent the model's capacity to generalise to other data partitions. There was no systematic hyperparameter optimisation, therefore the ANN architecture of 64 and 32 neurones per hidden layer was chosen heuristically rather than using grid search or Bayesian optimisation, and an alternative configuration may produce better results. Finally, the Cabin and Name elements were completely removed rather than re-engineered, potentially leaving predictive signal on the table.

5. Conclusion

This report successfully predicted passenger survival on the Titanic dataset using an Artificial Neural Network (MLP Classifier) together with Logistic Regression and Decision Tree baselines. Starting directly from the preprocessing pipeline established in Assignment 1 and expanding it with the removal of non-predictive identifier columns, the comprehensive pipeline included data cleaning, missing value imputation, categorical encoding, standard scaling, stratified train-test splitting, model training, and systematic performance analysis throughout five metrics.

The two goals of the study were accomplished. The comparative evaluation showed that the network's non-linear modelling capacity consistently outperforms both the linear Logistic Regression baseline and the non-linear Decision Tree baseline. The ANN was implemented and trained, achieving the highest accuracy and AUC-ROC among the three models evaluated (exact values in Table 2). The predominant predictive roles of passenger class and sex, the significance of feature scaling for stable ANN convergence, the False Negative bias that results from class imbalance, and the consistency of the results with those published in recent studies on the same dataset are some of the key findings.

Future research must focus on class weighting or decision threshold tuning to improve recall for the survived class, cross validation for more accurate performance estimates, and feature engineering techniques like creating a family size indicator and extracting titles from passenger names, which the literature indicates can push reliability above the 85% threshold representing the ceiling for models trained on the raw Titanic features only.

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